UniRS: Toward Unified Multitask Fine-Tuning for Remote Sensing Foundation Model

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***Abstract*— While the recent advancements highlight the significant potential of the remote sensing foundation model in addressing Earth observation tasks, they require fine-tuning with task-specific data to transfer to various applications. Fine-tuning for each task greatly limits the generalizability of foundation models. The process directs the models toward the task-relevant information with less focus on the universal feature extraction for multiple tasks, which is considered the primary advantage of foundation models. Moreover, it also overlooks the complementary relationships among tasks and significantly increases the computational complexity. However, simply training a model with multiple tasks also struggles to find effective representations of features. RSI interpretation tasks exhibit substantial fine-grained differences, with each task potentially favoring features at different scales. Additionally, using the same features for all tasks can cause mutual interference, and the lack of multitask datasets in remote sensing society hampers the generalization and sharing of visual features across tasks. To address these issues, we propose a multitask fine-tuning framework, *UniRS*. To facilitate cross-scale feature interactions, we introduce the cross-scale generic feature interaction (CGFI) mechanism, which integrates multiscale features and uses generic feature query (GFQ) to achieve more generalized representations. Additionally, we propose the task-specific low-rank mixture of experts (TLMoE) decomposition. General features are fed into multiple low-rank experts, each capable of capturing different patterns of the features. By dynamically combining the outputs of these experts, features for each sample and each task are encoded dynamically. Furthermore, we have expanded the existing datasets and developed MOTA, a new dataset annotated for three tasks: semantic segmentation (SS), rotated object detection, and multilabel scene classification (CLS). Extensive experiments demonstrate the effectiveness of the well-designed *UniRS*. The code and dataset will be released at https://github.com/Duckyee728/UniRS**

***Index Terms*— Fine-tuning, foundation model, object detection, remote sensing imagery (RSI), scene classification (CLS), semantic segmentation (SS).**

I. Introduction

ADVANCES in Earth observation techniques have led to generating a large volume of high-quality remote sensing imagery (RSI). These data are easily accessible, can be obtained in real time, and provide rich geospatial and spectral information, making them a significant source of information. With the rapid growth of deep learning, some fundamental RSI tasks, such as scene classification (CLS), rotated object detection (RD), and semantic segmentation (SS), play a vital role in applications of RSI interpretation [1], [2] and well-studied in the geomatics community. In previous works, researchers focused on carefully designing deep network architectures targeted at specific tasks, training these models with extensively annotated data. This approach has significantly booted the field of RSI interpretation, facilitating its practical application in various critical areas such as precision agriculture [3], [4], urban planning [5], [6], and natural hazards detection [7].

Despite the tremendous progress made by previous task-specific frameworks, these methods require significant data and computational resources to train multiple deep models when handling different tasks simultaneously. Remote sensing foundation models, which pretraining on large datasets to enhance their feature representation capabilities, have profoundly transformed research in remote sensing interpretation. Foundation models can greatly enhance the capabilities of various downstream tasks, reducing the need to train task-specific models from scratch and thereby improving the robustness of deep learning approaches. Wang et al. [8] explored remote sensing pretraining (RSP) using the MillionAID dataset. Subsequent works have focused on designing network architectures and pretraining strategies that are more suitable for RSI, aiming further to enhance the feature extraction capabilities of foundation models [9], [10], [11]. While these foundation models have shown excellent performance across various downstream tasks, to transfer these foundation models for specific downstream tasks, it is common practice to fine-tune these large-scale pretrained models for specific downstream tasks. In this study, we assume that the current extensively adopted "pretraining and fine-tuning" strategy has not reached the final and optimal goal of the foundation model due to the following observations.

1) *Task-Irrelevant Redundant Training:* RSI contains many task-independent features such as texture and color, resulting in redundant training when foundation models are fine-tuned independently for specific tasks.
2) *Lack of Multitask Feature Interaction:* Individual fine-tuning does not fully exploit the complementary nature of multiple tasks, limiting the foundation models' ability to capture essential feature representations for diverse remote sensing applications.

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3) *Increased Computational Complexity:* Fine-tuning for individual tasks significantly raises storage requirements and computational costs.

These insights have inspired our research into multitask fine-tuning of remote sensing foundation models, as shown in Fig. 1. It allows fine-tuning these models once they transfer to multiple downstream tasks, significantly reducing computational demands. Moreover, the shared features among tasks facilitate the sharing and interaction of domain knowledge across tasks, thereby enhancing the model's ability to tackle downstream challenges effectively. However, establishing a multitask fine-tuning framework using existing resources in the remote sensing society faces the following challenges.

1) Earth observation tasks vary greatly—for instance, SS, object detection, and scene CLS involve substantial fine-grained differences in labels. This variability can lead to mutual interference among tasks, hindering the learning of shared features.
2) Feature sharing among tasks may interfere with the fine-tuning process, especially pronounced in RSI multitask fine-tuning due to the substantial differences among tasks.
3) There is a lack of multitask RSI datasets that include various types of annotations.

To address these issues, we propose a multitask fine-tuning framework for the remote sensing foundation model, ***UniRS***. The proposed method is integrated between the backbone and the decoders for different tasks, facilitating the interaction among multiscale features in the backbone and generating task-specific features for different tasks, thereby enhancing multitask learning (MTL) performance. Specifically, the cross-scale generic feature interaction (CGFI) module is proposed to generate more generic features. The RSI perception models often rely on multiscale feature maps, while different tasks exhibit varying preferences for different scales. Intuitively, CLS models tend to favor global features as summaries of the images. Segmentation models, on the other hand, require high-resolution features to parse each pixel accurately. Based on these observations, CGFI integrates multiscale feature maps across the channel and spatial dimensions and employs the self-attention mechanism for feature interaction to achieve cross-scale semantic fusion. Subsequently, a generic feature query (GFQ) is designed to retrieve more effective general feature representations. To further generate task-specific features, the task-specific low-rank mixture of experts (TLMoE) decomposition is proposed. This module comprises multiple low-rank experts and a gating network. Features from the backbone are fed into different experts, which output distinct task-dependent representations. The gating network generates gating scores for each expert based on the task and weights them accordingly to produce task-specific features. Inspired by low-rank adaptation in the LLM domain, we use low-rank decomposition to reduce the experts' parameters and computational overhead. The proposed modules are plug-and-play and can be easily extended to accommodate any number of tasks. Moreover, we conduct a multitask RSI perception dataset, labeling all images with three types of annotations: multilabel scene CLS, rotated object detection, and SS, called MOTA. In summary, our contributions can be summarized as follows.

1) We revisit the pretraining and fine-tuning paradigm of foundation models in remote sensing society, exploring multitask fine-tuning. A multitask fine-tuning framework, ***UniRS***, is proposed to enhance the foundation model's ability to adapt to downstream tasks through the sharing and interaction of features across multiple tasks.
2) ***UniRS*** comprises CGFI and TLMoE decomposition. CGFI integrates and interacts with multiscale features to acquire more generic features. TLMoE dynamically generates task-specific features for each task using low-rank experts. Experiments have demonstrated the effectiveness of the proposed method.
3) We propose a multitask dataset, MOTA, which contains 15 000 RSI images, with each image annotated for three tasks: SS, rotational object detection, and multilabel scene CLS. Experiments show that multitask fine-tuning on MOTA effectively enhances the performance of each task.

II. Related Works

A. Foundation Models in Remote Sensing

The foundation model refers to the large-scale pretrained model designed to serve as a versatile starting point for various downstream tasks across different domains [12]. These models leverage extensive datasets and advanced training techniques (mostly self-supervised pretraining) to capture a wide range of knowledge and patterns, enabling them to perform effectively in diverse applications such as natural language processing and computer vision. Due to significant differences from natural images in terms of imaging perspectives and distances, spatiotemporal resolution, and spectral range, using models pretrained on natural images for remote sensing tasks can lead to suboptimal performance. Consequently, specialized remote sensing models and pretraining strategies have been developed to extract more effective representations of general remote sensing features. Seco proposed the seasonal contrast to utilize large-scale unlabeled remote sensing datasets [13]. SatMAE [14] used temporal and multispectral satellite imagery to pretrain Transformers. Wang et al. [9] proposed a new rotated varied-size window attention (RVSA) to enhance the performance of the plain vision Transformer and trained a 100 M parameters foundation model for RSI. HyperSIGMA [15] further explored the large-scale pretraining in hyperspectral images, and the first built the billion-level foundation model specifically. The development of foundation models for remote sensing has significantly advanced the performance of Earth observation tasks. However, current studies extensively adopted a "pretraining and fine-tuning" strategy while neglecting to improve the models' generalizability in the fine-tuning stage. In this work, we rethink the existing adaptation methods and encourage multitask fine-tuning, enabling the foundation model to adapt to multiple downstream tasks within a single fine-tuning process.

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B. Adapting Foundation Models

Fine-tuning is a process where a pretrained model is further trained on a specific task or dataset to adapt its parameters for improved performance in that particular context. This involves continuing the training process with a smaller learning rate, allowing the model to retain its generalized knowledge while optimizing it for specific objectives. Current studies primarily focus on reducing the number of fine-tuning parameters, thereby lowering training costs, by either freezing a portion of the foundation model's parameters or selectively training specific layers. LoRA [16] decomposed the weight updates into low-rank matrices. Instead of updating the entire weight matrix during fine-tuning, LoRA introduced two low-rank matrices A and B that approximate the update. An adapter-based method [17], [18], [19] aimed at task-specific adaptations without modifying the entire network. Adapter layers are small, and additional layers are inserted into a pretrained model to enable efficient fine-tuning. These layers are trained while keeping the original model parameters fixed. However, these fine-tuning methods typically focus on optimizing for a single specific task. When applied to multiple tasks, they often overlook potential intertask relationships and significantly increase training time. Based on this observation, we explore multitask fine-tuning of remote sensing foundation models.

C. Multitask Learning

The unique aspect of human vision is its ability to process multiple tasks within the same scene parallelly, such as object recognition, scene understanding, and motion prediction. This capability enables humans to make quick decisions in complex environments. MTL is an emerging trend to solve this problem [20], [21], which can avoid repetitive training of task-specific models, allowing for the sharing of model parameters and feature representation to generate predictions for different tasks with only one-time inference simultaneously. Recently, several studies have attempted to use MTL to enhance the performance of Earth observation tasks. MTP [11] utilizes the SAM to generate segmentation masks at the instance or semantic level using the existing object detection annotations and conducts a multitask pretraining pipeline. RSCoTr [22] explores the cotraining strategy of CLS, RD, and SS tasks in RSI in a tricky way, and three well-designed decoders are used, which makes simple extensions to more task number scenarios difficult. Inspired by the feature-sharing mechanism of MTL, this article explores multitask fine-tuning. Unlike existing studies, we focus on enhancing the foundation model's generalization during the fine-tuning stage.

III. Methodology

A. Framework Overview

In this section, we start with the definition of multitask fine-tuning. Given T tasks $t_1, t_2, \ldots, t_T$ (e.g., SS, object detection, and scene CLS), our dataset contains for each task t a set of training pairs $\{x, y_t\}^N$, where N denotes the number of the images. Note that the training image set x is identical, while the ground truth varies from tasks. Following the previous vision perception works, we employ a CNN-based or

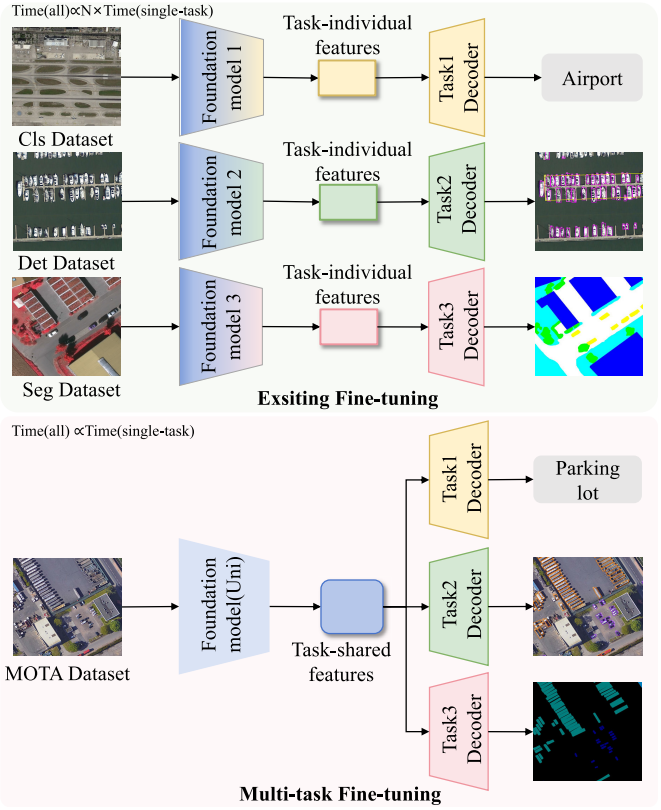


Fig. 1. Illustration of the existing fine-tuning methods and the multitask fine-tuning. The foundational model is pretrained on large datasets to acquire robust feature extraction capabilities. Existing fine-tuning methods require training a separate model for each task, resulting in inefficiencies in both time and storage. This article explores multitask fine-tuning, which enhances the performance of individual tasks through shared and interactive features across multiple tasks. Time$(\cdot)$ denotes the fine-tuning time.

Transformer-based foundation model $f(\cdot)$ to extract features. The image $x \in \mathbb{R}^{H\times W\times 3}$ is input to the backbone, generating four stages of image features, and then obtain predictions with task-specific decoders $g_t(\cdot)$. The overall objective loss can be formulated as

$$\mathcal{L} = \sum_{t=1}^{T} \lambda_t \cdot \mathcal{L}_t(g_t(f(x)), y_t) \tag{1}$$

where H, W, and 3 are the height, width, and channel of the image, respectively. Fig. 2 shows the overall illustration of our proposed framework. In proposed ***UniRS***, two well-designed blocks are plugged into the different stages of the backbone. To better utilize the scale information in features, CGFI employs space–channel mixing. It uses scale query to obtain more generalized task-generic features, which will be detailed in Section III-B. TLMoE decomposition leverages the mixture of experts (MoE) model, combining the characteristics of each task to generate task-specific features, which will be elaborated in Section III-C.

B. Cross-Scale Generic Feature Interaction

In this module, we first concatenate and fuse features from different layers to achieve information interaction among features of various scales. Subsequently, the task-specific features for the ith layer are obtained through scale

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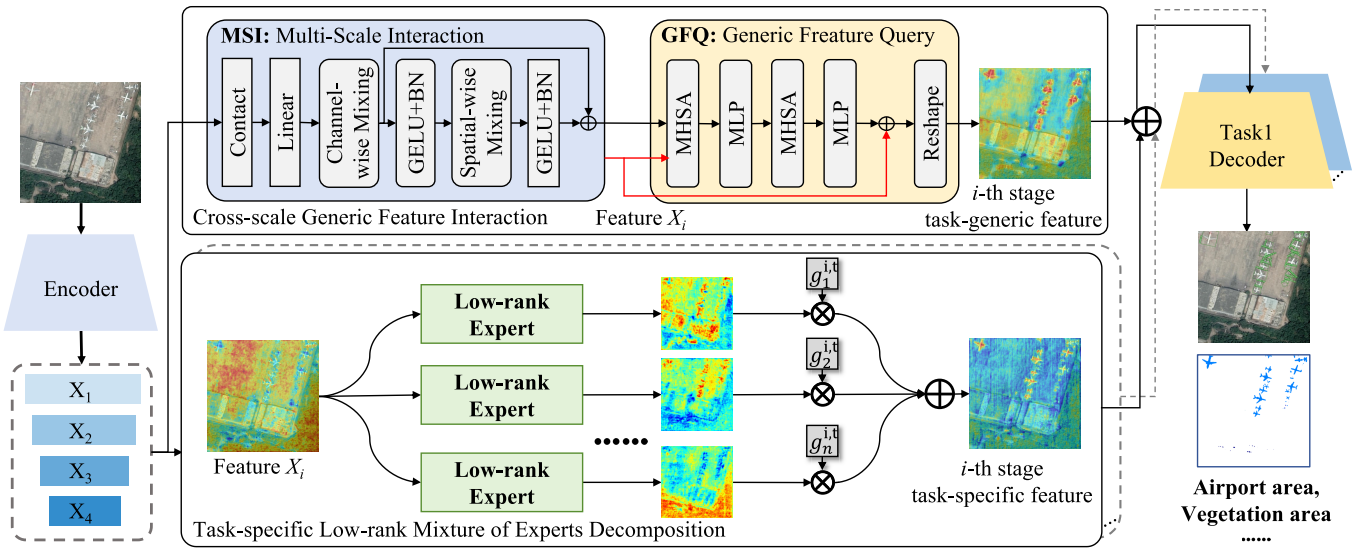


Fig. 2. Overview of the proposed ***UniRS***. It is integrated between the backbone and the decoder to enhance multitask interaction capabilities. Specifically, the proposed method comprises two components: CGFI and TLMoE decomposition. CGFI promotes feature interaction across different scales by utilizing a channel–spatial fusion mechanism. In parallel, TLMoE leverages an LMoE framework, dynamically assigning unique gating scores to each expert. These gating scores combine the outputs of various experts, producing task-specific features that are then passed into the task decoder.

information querying. Given the multiscale features extracted from the backbone, $[X_1, X_2, X_3, X_4]$, different visual tasks demonstrate varying preferences for features of different scales, presenting a challenge for MTL. We have modeled the relationships between features at various scales by integrating and interacting across scales to achieve more generalized multiscale information. To enable the information interaction between different scales, we first interoperate all the features into the same shape $X_i \in \mathbb{R}^{(H/4)\times(W/4)\times C}$ and contact them along the channel dimension, which can be formulated as $X' \in \mathbb{R}^{(H/4)\times(W/4)\times 4C}$. A linear layer is equipped to reduce the hidden dimension of the feature and get the input feature map $X_{C'} \in \mathbb{R}^{(H/4)\times(W/4)\times C'}$. Inspired by [23], we utilized standard point-wise convolution (with a 1×1 kernel) to perform channel-wise and spatial-wise mixing on the stacked multiscale features, thereby promoting communication between features of different scales. Suppose that W_1 and b are the point-wise convolution weight and learnable bias, respectively, and the channel-wise mixing can be termed as

$$X_{C'} = \sum_{C'=0}^{C'-1} W_1 \cdot X_{C'} + b. \tag{2}$$

Then, a GELU activation and a BatchNorm are added, which can be formulated as

$$X_{C'} = \mathrm{BN}(\sigma(X_{C'})) \tag{3}$$

where $\sigma(\cdot)$ is the GELU and BN represents the BatchNorm operation. To model spatial interactions, image feature $X_{C'}$ is fed into a convolution to learn the corresponding offsets $\Delta_{(i,j)}$, generating the relative offsets concerning the reference point. For each location point (i, j) on the feature, the spatial mixing can be formulated as

$$D_S(X_{i,j}) = \sum_{C'=0}^{C'-1} W_2 \cdot X\big((i, j) + \Delta_{(i,j)}, C'\big) \tag{4}$$

where W_2 is a deformable weight, and the offset $\Delta_{(i,j)}$ is the learnable offset. After spatial-wise mixing, a GELU activation, BatchNorm, and residual connection are applied

$$X_{C'} = X_{i,j} + \mathrm{BN}\big(\sigma\big(D_S\big(X_{i,j}\big)\big)\big). \tag{5}$$

To facilitate the next operation, we flatten the feature $X_{C'} \in \mathbb{R}^{(H/4)\times(W/4)\times C'}$ to a sequence $X_s \in \mathbb{R}^{N\times C'}$ $(N = (H/4) \times (W/4))$. The sequence integrates information from multiple scales and ensures comprehensive mixing and interaction across both channel and spatial dimensions. Subsequently, the proposed method employs the attention mechanism to capture scale information, thereby obtaining more generalized multiscale features. Specifically, we utilize the multihead self-attention (MHSA) to conduct GFQ. The self-attention operator can be denoted as

$$\mathrm{MHSA}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{6}$$

where $Q \in \mathbb{R}^{N\times C'}$, $K \in \mathbb{R}^{N\times C'}$, and $V \in \mathbb{R}^{N\times C'}$ are the query, key, and value matrices, $\mathrm{MHSA}(Q, K, V) \in \mathbb{R}^{N\times \tilde{C}'}$, respectively. For all multiscale features X_i, we first reshape it to a sequence and treat them as the query using the MHSA to get the more generic feature from the mixed sequence X_s

$$X_i^g = \mathrm{MHSA}(Q = \mathrm{LN}(X_i)K = \mathrm{LN}(X_s), V = \mathrm{LN}(X_s)) \tag{7}$$

where the LN denotes the LayerNorm operation. After passing through the MLP module, we perform the MHSA operation again using different values as queries to enhance feature interaction further

$$X_i^g = \mathrm{MLP}\big(X_i^g\big) \tag{8}$$

$$X_i^g = \mathrm{MHSA}\big(Q = \mathrm{LN}\big(X_i^g\big), K = \mathrm{LN}\big(X_i^g\big), V = \mathrm{LN}\big(X_i^g\big)\big). \tag{9}$$

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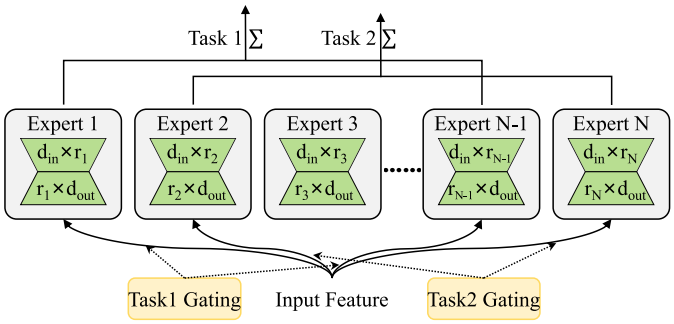


Fig. 3. Illustration of the proposed LMoE.

Note that the MLP module consists of a linear layer and a LayerNorm. Then, the generic feature of the ith stage is calculated as

$$\hat{X}_i = \text{Reshape}\left(X_i^g + \text{MLP}\left(\hat{X}_i\right)\right) \tag{10}$$

where Reshape denotes the rearrange operation, restoring the sequence to the corresponding feature scale.

C. TLMoE Decomposition

Different tasks often have varied preferences for features, and it is intuitive to generate task-specific features for each task. This allows the decoders for different tasks to focus more effectively on the features most beneficial for their respective tasks. In this section, we introduce a task-specific feature decoupling module that employs the low-rank mixture of experts (LMoE) to obtain distinctive task-specific features.

As illustrated in Fig. 2, to learn distinct task-specific representations from backbone features, we employed the MoE network. Formally, for each X_i at the ith stage of the backbone, suppose that MoE contains N experts and T router networks, denoted as $E = \{E_1, E_2, \ldots, E_N\}$ and $G = \{G_1, G_2, \ldots, G_T\}$, respectively. N and T are the number of experts and the number of tasks, respectively. Feature X_i is fed to the different experts, and these networks decompose the backbone features into several representative task-agnostic features within subfeature spaces, with each feature capturing a different pattern of the backbone characteristics. Simultaneously, the MoE utilizes task-specific router networks to learn gates for different tasks. By automatically combining these representative features, it derives task-specific features. For a given task t, the gate value for each expert generated by the router network is expressed as $\mathbf{g}^t = G^t(X)$. MoE generates the task-specific feature X^t for task t by utilizing gates to combine the expert features, which can be formulated as

$$X_i^t = f_{\text{moe}}(X_i) = \sum_{n=1}^{N} \mathbf{g}_n^{i,t} E_n(X_i) \tag{11}$$

where $f_{\text{moe}}(\cdot)$ denotes the MoE network and $E_n(\cdot)$ represents the nth expert. The expert and router network details, as shown in Fig. 3, will be illustrated next.

1) Low-Rank MoE: In previous successful applications of the MoE model, experts are often composed of one or more layers of MLP. These approaches can harness the diversity of multiple experts for feature encoding, while also introducing higher parameter and computational costs. In this section, Low-rank decomposition adaptation has been widely researched recently for efficiently fine-tuning pretrained models to accommodate various tasks. Inspired by LoRA [16], we replaced linear layers with convolutional layers in the form of low-rank decomposition. The matrix operation of the linear layer in experts can be described as follows:

$$O_E = \Delta W x + b \tag{12}$$

where $\Delta W \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ denotes the parameter matrix of the linear layer and O_E denotes the output of an expert. To improve the efficiency of the training process and conserve resources, we replaced the experts' parameter matrix with low-rank decomposition, which can be written as

$$\Delta W_E = BA \tag{13}$$

where $A \in \mathbb{R}^{d_{\text{in}} \times r_n}$ and $B \in \mathbb{R}^{r_n \times d_{\text{out}}}$, the rank $r_n \ll \min(d_{\text{in}}, d_{\text{out}})$. In the proposed LMoE, experts are assigned different r_n's to enhance the diversity of their feature outputs.

2) Router Network: After decoupling the backbone features using various low-rank experts to obtain several representative features, our goal is to automatically assemble these representative features to decode the discriminative, task-specific features for different tasks. To achieve this, a well-designed routing network is suggested. Given the ith stage feature form the backbone $X_i \in \mathbb{R}^{H \times W \times C}$, we use a 3×3 convolution to reduce its channel dimensions to $X_i' \in \mathbb{R}^{H \times W \times (C/4)}$. After using Reshape to convert the features into a sequence, MHSA is also used to extract discriminative information

$$s_i^t = \text{MHSA}\left(Q = \text{LN}\left(X_i'\right), K = \text{LN}\left(X_i'\right), V = \text{LN}\left(X_i'\right)\right). \tag{14}$$

After this, an MLP is equipped to map the channel dimension from $(C/4)$ to N. Drawing inspiration from [24] and [25], we introduced a parallel position-aware branch to enhance the gated network's ability to model long-range spatial context. Specifically, two MLPs are connected sequentially: the first is mapping the feature shape from $\mathbb{R}^{HW \times C}$ to $\mathbb{R}^{1 \times C}$ along the spatial dimensions and the second MLP further reduces to $\mathbb{R}^{1 \times N}$. The outputs from the MHSA and position-aware branches will be stacked along the channel dimension and then passed through a convolution to obtain the final task-specific routing features $F_i^t \in \mathbb{R}^{1 \times N}$. To sum up, the task gating scores of the tth task and the ith stage, i.e., $\mathbf{g}_n^{i,t} = \{\mathbf{g}_1^{i,t}, \mathbf{g}_2^{i,t}, \ldots, \mathbf{g}_N^{i,t}\}$, can be calculated as

$$\mathbf{g}_n^{i,t} = \text{TopK}\left(\text{Softmax}\left(F_i^t\right), K\right) \tag{15}$$

and we use the TopK function to deactivate some experts based on the corresponding gating scores

$$\text{TopK}(\boldsymbol{v}, K) = \begin{cases} \boldsymbol{v}, & \text{if } \boldsymbol{v} \text{ is in the top } K \text{ elements} \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

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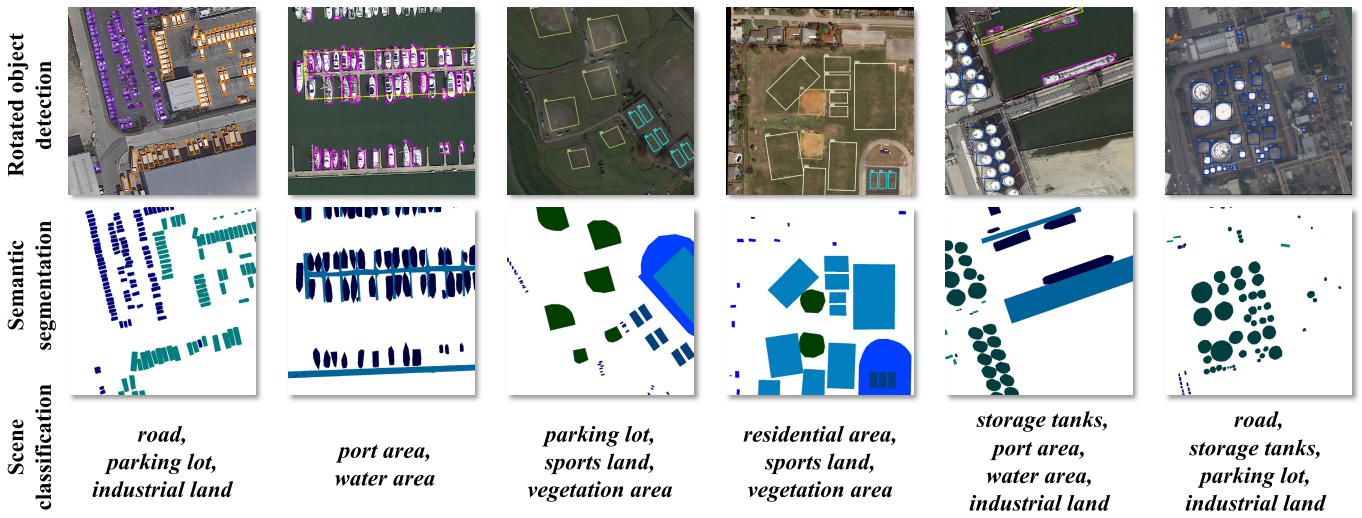


Fig. 4. Example display of the proposed MOTA.

The advantage of LMoE lies in its ability to dynamically encode the features for each task for each sample with lower resource consumption. Finally, combining the task-generic features generated by CGFI described in Section III-B, the output of the features to the decoder for the tth task can be represented as

$$\hat{X}_i^t = \hat{X}_i + \sum_{n=1}^{N} \mathbf{g}_n^{i,t} E_n(X_i). \tag{17}$$

Let $\hat{X}^t = \{\hat{X}_1^t, \hat{X}_2^t, \ldots\}$, and the total loss of ***UniRS*** is

$$\mathcal{L} = \sum_{t=1}^{T} \lambda_t \cdot \mathcal{L}_t\left(g_t\left(\hat{X}^t\right), y_t\right). \tag{18}$$

IV. Experiments

A. Dataset Construction

To enhance the multitask perception capabilities of RSI, we have expanded the DOTA [26] dataset by adding multilabel scene CLS labels, named MOTA, as shown in Fig. 4. DOTA is a large-scale dataset for object detection in RSI, consisting of 2806 images and 188 282 instances. Zamir et al. [27] expanded the dataset and provided SS labels for the original images. We utilize the rotate detection annotations provided by DOTA V1.0, and the semantic masks, to obtain the CLS ground truth, we split the images into the 1024 × 1024 patches following the previous works [28] in the single-scale setting with the corresponding masks. Due to the limitations of the DOTA dataset, we only process the *train* and *val* sets, which consist of 9734 and 5297 images, respectively, after splitting. We then annotate them. Given the high resolution of the images, a single patch may contain multiple scenes; therefore, we opt for multilabel annotation. This means that one image might encompass several scene categories. The number of all candidate scene labels is 13: road (RD), storage tanks (STs), port area (PA), runway (RW), parking lot (PL), airport area (AP), arable land (AL), residential area (RA), sports land (SL), vegetation area (VA), water area (WA), industrial land (IL), and cloud (CL). The co-occurrence matrix of labels in the CLS task is shown in Fig. 5.

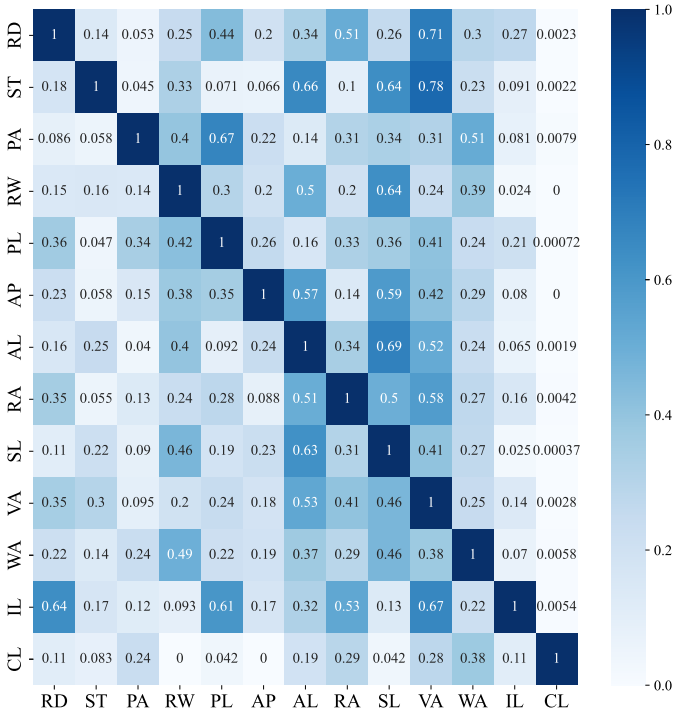


Fig. 5. Co-occurrence matrix of labels in CLS task in the MOTA dataset.

B. Implementation Details

We selected multilabel scene CLS, rotated object detection (RD), and SS as our target tasks, i.e., $T = 3$. We built the multitask RSI fine-tuning framework based on MMDetection and conducted all experiments on this foundation. To verify the applicability of the proposed method to different backbone networks, we chose ResNet-50 [29], InternImageXL [30], Swin-T [31], and Swin-L as the backbone networks. These choices cover both CNN- and Transformer-based architectures,

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encompassing a range of parameter scales. All of these backbones are initialized with pretrained weights on ImageNet. Following [11], we use the UperNet [32] as the segmentation decoder, Oriented R-CNN [33] as the object detection decoder, and a plain sigmoid classifier for CLS. The SGD optimizer was selected for other backbones, with an initial learning rate of 0.02, a weight decay of 0.0005, and a momentum of 0.9. The input image size during training is set to 1024 × 1024, and the batch size was configured to 8. All experiments were conducted on the PyTorch platform using four A100 SXM4 40-GB GPUs and were trained for 48 epochs.

Metrics: For CLS, we use the micro-average F_1 score (mF_1) to evaluate the classifiers' performance. Unlike traditional single-label CLS, where each image is assigned only one label from a set of mutually exclusive labels, multilabel CLS allows each image to belong to multiple categories simultaneously. mF_1 aggregates the contributions of all classes to compute the average F_1 score globally by counting the total true positives (TPs), false negatives (FNs), and false positives (FPs), calculated as follows:

$$F_1 = \frac{2p_e r_e}{p_e + r_e} \tag{19}$$

$$p_e = \frac{\mathrm{TP}_e}{\mathrm{TP}_e + \mathrm{FP}_e}, r_e = \frac{\mathrm{TP}_e}{\mathrm{TP}_e + \mathrm{FN}_e} \tag{20}$$

where TP_e, FP_e, and FN_e indicate the numbers of positive labels, which are TPs, FPs, and negative labels. To evaluate the SS performance, we adopt the most commonly used metric: mean intersection over union (mIoU), which is also computed based on the TP, true negative (TN), FP, and FN values

$$\mathrm{IoU}_l = \frac{\mathrm{TP}_l}{\mathrm{TP}_l + \mathrm{FN}_l + \mathrm{FP}_l} \tag{21}$$

$$\mathrm{mIoU} = \frac{1}{L}\sum_{l=1}^{L} \mathrm{IoU}_l \tag{22}$$

where IoU_l refers to category l. For RD, the AP of the lth class AP_l is the area under the P–R curve of the class $p_c(R)$, and we calculate the AP and mAP for evaluation

$$\mathrm{AP}_l = \int_0^1 p_l(R)\mathrm{d}R, \quad \mathrm{mAP} = \frac{1}{L}\sum_{l=1}^{L} \mathrm{AP}_l. \tag{23}$$

In this article, we selected two baselines: the single-task baseline, which represents fine-tuning on the respective individual task datasets, and the multitask baseline, which involves multitask adapting using the MOTA dataset. To evaluate the MTL performance of the proposed method, we computed the average relative difference of all tasks compared to the single-task baseline, referred to as multitask performance (MTL Perf Δ_m) [23]

$$\Delta_m = \frac{1}{T}\sum_{i=1}^{T}\left(F_{m,i} - F_{s,t}\right)/F_{s,t} \times 100\% \tag{24}$$

When Δ_m is higher, the performance gain of multi-task learning over single-task learning increases, reflecting the method's multi-task learning capability is better.

TABLE I

COMPARISON OF THE BASELINE WITH THE PROPOSED UNIRS ON VAL SET OF THE MOTA DATASET. THE NOTATION ↑: HIGHER IS BETTER. THE PARAMS DENOTE MODELS' PARAMETERS

Method	Backbone	Params (M)	CLS (mF_1) ↑	SS (mIoU) ↑	RD (mAP)↑	Δ_m [%] ↑
Single-task baseline	ResNet-50	103.22	80.32	63.18	73.32	0
Multi-task baseline		49.51	80.98	64.45	73.54	1.0
UniRs(Ours)		65.78	**82.54**	**65.45**	**73.91**	**2.4**
Single-task baseline	Swin-T	113.40	81.63	63.46	74.21	0
Multi-task baseline		52.91	82.76	64.32	74.92	1.2
UniRs(Ours)		69.19	**83.36**	**66.10**	**76.37**	**3.0**
Single-task baseline	Swin-L	616.95	82.13	65.80	76.16	0
Multi-task baseline		220.75	82.85	65.97	77.20	0.8
UniRs(Ours)		237.03	**84.00**	**67.56**	**77.55**	**2.3**
Single-task baseline	InternImage-XL	1018.81	81.75	66.31	76.37	0
Multi-task baseline		354.73	82.99	67.62	77.88	1.8
UniRs(Ours)		371.01	**83.42**	**68.04**	**78.41**	**2.4**

TABLE II

RESULTS OF THE MULTILABEL CLS MODEL WITH DIFFERENT BACKBONES AND SOTA METHODS ON THE VAL SET OF THE MOTA DATASET

Method	Backbone	mF1
SNDL [34]	ResNet-50	80.9
C-Tran [35]	ResNet-101	81.9
SIGNA [36]	ResNet-101	82.1
-	ResNet-50	80.3
-	Swin-T	81.6
-	Swin-L	82.1
-	InternImage-XL	81.7
-	ResNet-50+***UniRS***	82.5
-	Swin-T+***UniRS***	83.3
-	Swin-L+***UniRS***	**84.0**
-	InternImage-XL+***UniRS***	83.4

C. Comparison With the Baselines

As shown in Table I, compared to training task-specific models for different tasks separately (single-task baseline), multitask fine-tuning using the proposed MOTA dataset (multitask baseline) significantly enhances the performance of each task. It substantially reduces both training time and the number of model parameters. By sharing a feature extraction backbone network, multitask fine-tuning can exploit common features across tasks, improving performance for each task. Specifically, compared to the single-task baseline, the multitask baseline shows an average performance improvement of 0.8%–1.8% on each task. Moreover, the proposed ***UniRS*** further improves the performance of multitask fine-tuning, achieving the best performance for each task with each backbone. Due to the introduction of LMoE, our method incorporates a considerable number of experts at a small cost of increased parameters. We also observe that the proposed method performs even better under experimental settings with lower parameter backbones, achieving average improvements of 2.4% and 3.0% with ResNet-50 and Swin-T, respectively. InternImage-XL, due to its largest parameter count and strongest feature extraction capabilities, achieves the best performance on all tasks, with an improvement of 2.4% in Δ_m.

D. Comparison With Task-Specific Methods

1) Compare With the Multilabel Scene CLS Methods: Table II presents the results of the performance on the

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TABLE III
RESULTS OF THE UPERNET SEGMENTATION MODEL WITH DIFFERENT BACKBONES AND DIFFERENT METHODS ON THE VALIDATION SET OF THE ISAID DATASET

Decoder	Backbone	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIOU
FCN [37]	VGG-16	51.7	22.9	26.4	74.8	30.2	27.9	8.2	49.3	37.0	0	30.7	51.9	52.1	62.9	42.0	41.7
PSPNet [38]	ResNet-50	65.2	52.1	75.7	85.6	61.1	60.2	32.5	58.0	43.0	10.9	46.8	68.6	71.9	79.5	54.3	60.3
DeeplabV3+ [39]	ResNet-50	63.9	52.5	72.8	84.9	56.5	58.9	32.2	59.1	42.9	31.4	46.1	67.7	72.9	79.8	52.6	60.8
CCNet [40]	ResNet-50	61.4	65.7	68.9	82.9	57.1	56.8	34.0	57.6	38.3	31.6	36.5	57.2	75.0	75.8	45.9	58.3
HRNet [41]	HRNetW-18	65.9	68.9	74.0	86.9	59.4	61.5	33.8	62.1	46.9	14.9	44.2	52.9	75.6	81.7	52.2	61.5
RANet [42]	ResNet-50	67.1	61.3	72.5	85.1	53.2	47.1	45.3	60.1	49.3	38.1	41.8	70.5	58.8	83.1	55.6	62.1
AlignSeg [43]	ResNet-50	67.4	68.9	76.2	86.2	62.1	52.0	28.7	60.7	50.3	31.2	45.7	56.2	71.2	82.9	54.8	62.1
OCR [44]	HRNet-W48	67.8	70.7	73.6	87.9	63.4	47.7	33.1	61.4	49.6	30.4	48.4	59.5	72.8	83.3	53.3	62.6
HMANet [45]	ResNet-50	65.4	70.9	74.7	88.7	60.5	54.6	29.0	59.7	50.3	32.6	51.4	62.9	70.2	83.8	51.9	62.6
FarSeg [46]	ResNet-50	65.4	61.8	77.7	86.4	62.1	56.7	36.7	60.6	46.3	35.8	51.2	71.4	72.5	82.0	53.9	63.7
FactSeg [47]	ResNet-50	68.3	56.8	78.4	88.9	64.9	54.6	36.3	62.7	49.5	42.7	51.5	69.4	73.6	84.1	55.7	64.8
RSSFormer [48]	RSS-B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	65.9
UperNet [32]	ResNet-50 + SeCo [13]	63.9	71.7	66.9	69.9	54.5	45.9	38.9	58.2	44.8	33.2	9.3	52.3	71.6	83.3	51.4	57.2
UperNet	ResNet-50+RSP [8]	64.2	75.9	68.8	69.9	58.5	54.4	40.2	59.6	47.5	32.1	43.8	65.4	76.5	82.8	51.5	61.6
UperNet	Swin-T+RSP [8]	67.0	74.6	73.7	70.7	59.0	60.1	**44.3**	62.0	50.6	37.6	46.8	64.9	76.2	85.2	53.8	64.1
UperNet	ViTAEv2-S	71.4	**77.5**	68.2	71.0	60.8	61.9	43.0	63.8	**53.6**	**43.4**	44.8	65.1	77.9	**86.4**	57.7	65.3
UperNet	Swin-B + RingMo [10]	73.5	73.0	77.0	89.1	62.5	**63.0**	43.2	63.9	51.2	40.1	48.9	67.3	**78.5**	85.7	58.9	67.2
UperNet	ResNet-50	63.7	55.2	73.1	89.3	68.5	52.5	42.1	62.5	47.2	29.9	45.7	72.4	74.4	78.8	56.6	63.2
UperNet	Swin-T	65.5	62.4	76.8	87.7	59.0	51.5	36.5	62.5	47.3	40.6	44.9	72.1	71.9	80.9	56.8	63.5
UperNet	Swin-L	68.5	66.0	78.6	88.5	63.0	58.4	40.6	65.0	47.2	46.5	45.4	73.0	73.0	82.4	58.0	65.8
UperNet	InternImage-XL	72.2	57.5	79.4	89.8	67.2	59.4	37.7	66.7	52.4	42.3	46.3	74.2	72.9	84.0	59.8	66.3
UperNet	ResNet-50 + ***UniRS***	67.6	70.0	72.2	90.4	71.3	47.7	41.6	66.5	49.3	34.5	48.8	72.6	74.8	81.1	59.8	65.5
UperNet	Swin-T + ***UniRS***	69.4	54.3	79.8	90.0	68.5	52.8	43.6	67.7	50.3	39.7	**51.0**	74.7	74.1	82.9	60.0	66.1
UperNet	Swin-L + ***UniRS***	72.0	60.1	**83.1**	90.3	71.2	57.8	41.6	68.2	50.9	42.1	48.8	74.6	77.4	84.1	60.0	67.6
UperNet	InternImage-XL + ***UniRS***	**73.2**	66.0	81.8	**90.6**	**71.6**	58.4	37.9	**68.7**	52.2	43.2	47.3	**75.6**	74.5	85.3	**63.3**	**68.0**

[1] PL: plane. BD: baseball diamond. BR: bridge. GTF: ground track field. SV: small vehicle. LV: large vehicle. SH: ship. TC: tennis court. BC: baseball court. ST: storage tank. SBF: soccer ball field. RA: roundabout. HA: harbor. SP: swimming pool. HC: helicopter.

multilabel scene CLS task in MOTA datasets. We employed a simple linear layer combined with a sigmoid function as a multilabel classifier for our proposed method. Compared to more complex, specially designed multilabel classifiers, ***UniRS*** demonstrates significant performance improvements with the same or fewer parameters in the backbone. Specifically, when using the same backbone, ***UniRS*** outperforms the SNDL classifier by 1.6%. Furthermore, classifiers utilizing ResNet-101 as the backbone perform worse than ResNet-50 trained with ***UniRS***. Across all backbone configurations, training with ***UniRS*** significantly enhances the task performance compared to the baseline.

2) Compare With the SS Methods: The SS part of the MOTA dataset utilizes the labels provided by iSAID [27]. We trained our models using the training set and reported performance on the validation set for a comprehensive comparison. As shown in Table III, we selected several single-task SS models for benchmarking. Our approach employs UperNet as the segmentation framework, and it was evident that using ***UniRS*** significantly improved SS performance while retaining the ResNet-50 backbone. Additionally, with Swin-T as the backbone, the proposed method still boosted model performance. Specifically, compared to RSP pretraining, our method achieved a 2.0% improvement and a 2.6% improvement over the IMP baseline when equipped with Swin-T as the backbone. Using larger backbones, such as InternImage-XL and Swin-L, ***UniRS*** enabled the models to reach state-of-the-art segmentation performance. Specifically, compared to the highest benchmarking method RingMo, ***UniRS*** with Swin-L and InternImage-XL achieved increases in mIoU by 0.4% and 0.8%, respectively. Even when compared with other methods using different decoders, the proposed method still demonstrated an advantage. This highlights the beneficial impact of multitask fine-tuning on enhancing single-task learning capabilities and the effectiveness of the proposed ***UniRS***.

3) Compare With the Rotated Object Detection Methods: Since the MOTA dataset directly uses the rotated object detection labels from the DOTA V1.0 dataset, this article's experimental setup is consistent with the single-scale condition of DOTA V1.0. To conduct a broader comparison, we trained the proposed method using the train and val sets of MOTA and then predicted the images of DOTA's test set. The experimental results were obtained by submitting the predictions for online testing, as shown in Table IV. Compared to other methods, using ***UniRS*** for training with a ResNet-50 network achieves performance equivalent to that of using a ResNet-101 as the backbone when the Oriented R-CNN is also used as the detector. Compared to other decoders with ResNet-50 or Swin-T backbones, these experimental setups also achieved competitive results. Using InterImage-XL as the backbone, the mAP can reach 80.22%, which is the state-of-the-art performance on this dataset under single-scale conditions.

E. Ablation Works

In this section, we conduct ablation experiments on various components of the proposed method. All experiments are conducted using ResNet-50 as the backbone and are trained on the MOTA dataset, and the performance of the validation set is reported.

1) Component Analysis: The proposed ***UniRS*** includes CGFI, which generates more distinctive multiscale features, and TLMoE decomposition, which utilizes the various low-rank experts to generate task-dependent features for each task. We evaluate their effectiveness in Table V through

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TABLE IV
RESULTS OF DIFFERENT METHODS ON THE TESTING SET OF THE DOTA-V1.0 DATASET

Decoder	Backbone	PL[1]	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
ROI Trans. [49]	ResNet-101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
R^3Det [50]	ResNet-101	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
Gilding Vertex [51]	ResNet-50	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
AOPG [52]	ResNet-101	89.14	82.74	51.87	69.28	77.65	82.42	88.08	90.89	86.26	85.13	60.60	66.30	74.05	67.76	58.77	75.39
DODet [53]	ResNet-101	89.61	83.10	51.43	72.02	79.16	81.99	87.71	90.89	86.53	84.56	62.21	65.38	71.98	70.79	61.93	75.89
S^2ANet [54]	ResNet-101	88.70	81.41	54.28	69.75	78.04	80.54	88.04	90.69	84.75	86.22	65.03	65.81	76.16	73.37	58.86	76.11
ReDet [55]	ReResNet-50	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
R^3Det-KLD [56]	ResNet-50	88.90	84.17	55.80	69.35	78.72	84.08	87.00	89.75	84.32	85.73	64.74	61.80	76.62	78.49	70.89	77.36
Ori. RepP. [57]	Swin-T	89.11	82.32	56.71	74.95	80.70	83.73	87.67	90.81	87.11	85.85	63.60	68.60	75.95	73.54	63.76	77.63
AO2-DETR [58]	ResNet-50	89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.68	72.39	71.62	77.73
Oriented R-CNN [33]	ResNet-50	89.58	81.35	54.86	71.35	79.04	83.03	88.16	90.90	87.43	84.91	63.39	66.67	74.19	69.05	58.14	76.14
Oriented R-CNN	Swin-T +RSP [8]	89.54	84.84	52.91	74.50	78.96	84.02	87.83	90.86	85.90	84.84	62.90	67.33	74.45	70.61	57.36	76.12
Oriented R-CNN	ResNet-101	88.86	83.48	55.27	76.92	74.27	82.10	87.52	90.90	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
Oriented R-CNN	ViTAE-B	89.40	80.02	55.25	70.90	80.84	85.45	88.20	90.87	83.56	86.39	56.54	69.87	77.11	79.63	67.47	77.43
Oriented R-CNN	ViT-B + RVSA [9]	89.23	81.44	57.91	72.94	79.91	85.08	88.23	90.87	87.37	86.68	59.07	**73.62**	77.12	78.70	71.06	78.61
Oriented R-CNN	ViTAE-B + RVSA [9]	89.38	84.26	**59.39**	73.19	79.99	85.36	88.08	90.87	88.50	86.53	58.93	72.24	77.31	79.59	71.24	78.99
Oriented R-CNN	ResNet-50 + ***UniRS***	89.46	79.98	54.18	71.56	74.38	83.38	88.07	90.89	87.07	85.35	59.55	66.54	76.18	72.59	65.12	76.29
Oriented R-CNN	Swin-T + ***UniRS***	89.60	83.32	55.14	75.27	79.07	84.55	88.08	**90.91**	86.31	86.64	62.33	66.22	75.29	71.94	70.55	77.68
Oriented R-CNN	Swin-L + ***UniRS***	89.85	**85.79**	54.51	77.66	79.14	85.30	88.24	90.90	**87.37**	87.20	66.92	69.51	76.06	71.62	**77.64**	79.18
Oriented R-CNN	InternImage-XL + ***UniRS***	**89.92**	85.47	58.64	**78.23**	**80.92**	**85.74**	**88.45**	90.90	86.20	**87.56**	**67.14**	69.77	**78.03**	**79.63**	76.76	**80.22**

[1] PL: plane. BD: baseball diamond. BR: bridge. GTF: ground track field. SV: small vehicle. LV: large vehicle. SH: ship. TC: tennis court. BC: baseball court. ST: storage tank. SBF: soccer ball field. RA: roundabout. HA: harbor. SP: swimming pool. HC: helicopter.

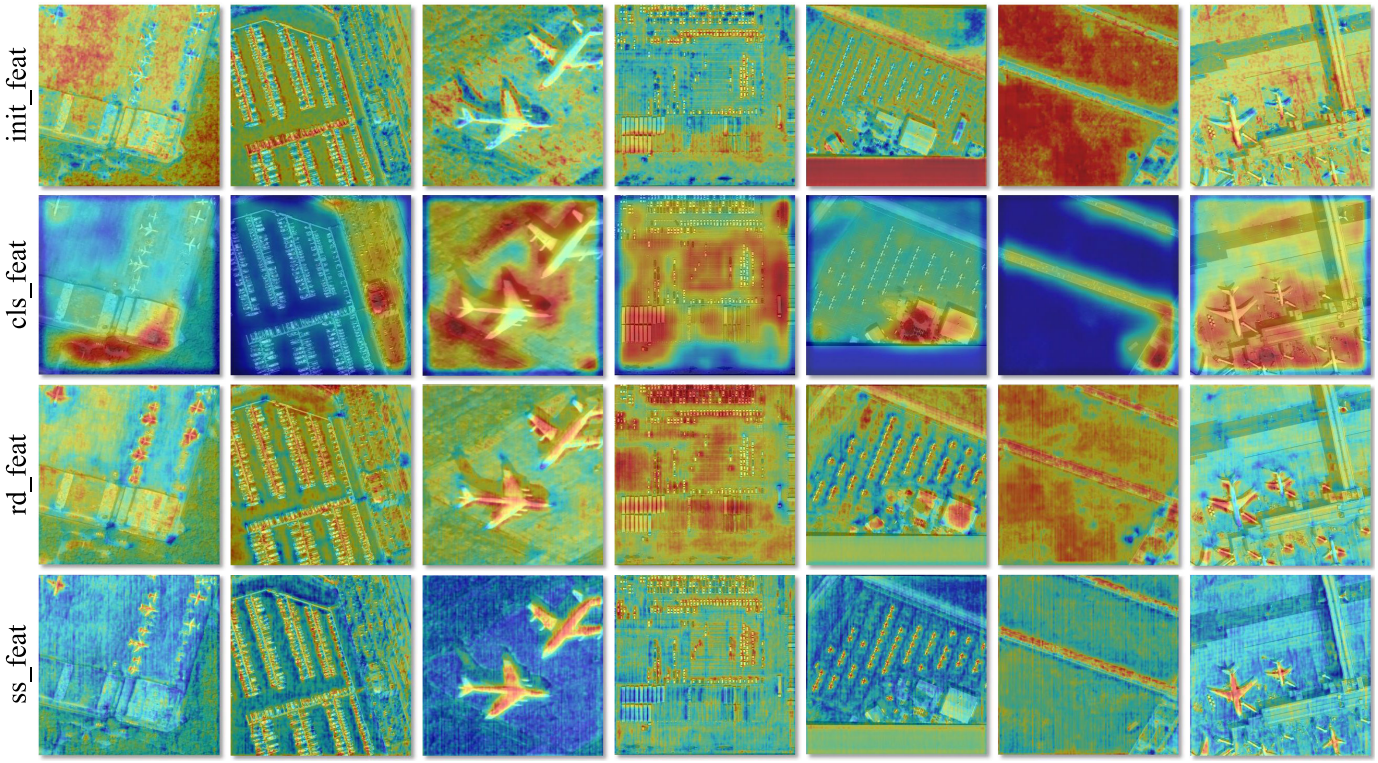


Fig. 6. Visualized feature of the ***UniRS*** output. Init_feat represents the features fed into different encoders without using ***UniRS***; cls_feat, rd_feat, and ss_feat, respectively, denote the task-specific features generated through ***UniRS***'s decoupling of generalized features, which are subsequently fed as input to the CLS, RD, and SS decoders.

TABLE V
EFFECTIVENESS OF THE PROPOSED COMPONENTS ON COMPARISON OF THE BASELINE WITH THE PROPOSED UNIRS ON VALIDATION SET OF THE MOTA DATASET

MTL	CGFI	TLMoE	CLS(mF_1)	SS(mIoU)	RD(mAP)	Δ_m [%]
-	-	-	80.32	63.44	73.75	0
✓	-	-	80.98	64.45	73.54	1.0
✓	✓	-	81.43	64.82	73.43	1.4
✓	-	✓	81.67	65.21	73.52	1.7
✓	✓	✓	**82.54**	**65.45**	**73.91**	**2.4**

detailed ablation studies. The MTL denotes using the MOTA dataset to conduct multitask fine-tuning. Compared to the single-task baseline, multitask fine-tuning brought about a 1.0% improvement in average task performance. The implementation shows that both proposed modules can further enhance the MTL performance. Specifically, CGFI and TLMoE improve the multitask baseline by 0.4% and 0.7%, respectively, while combining the two modules led to a further significant performance boost. Comparatively, TLMoE proved to be more effective, likely because this strategy greatly expands the way features are encoded and generates independent features for different tasks, thereby reducing interference from joint MTL to some extent.

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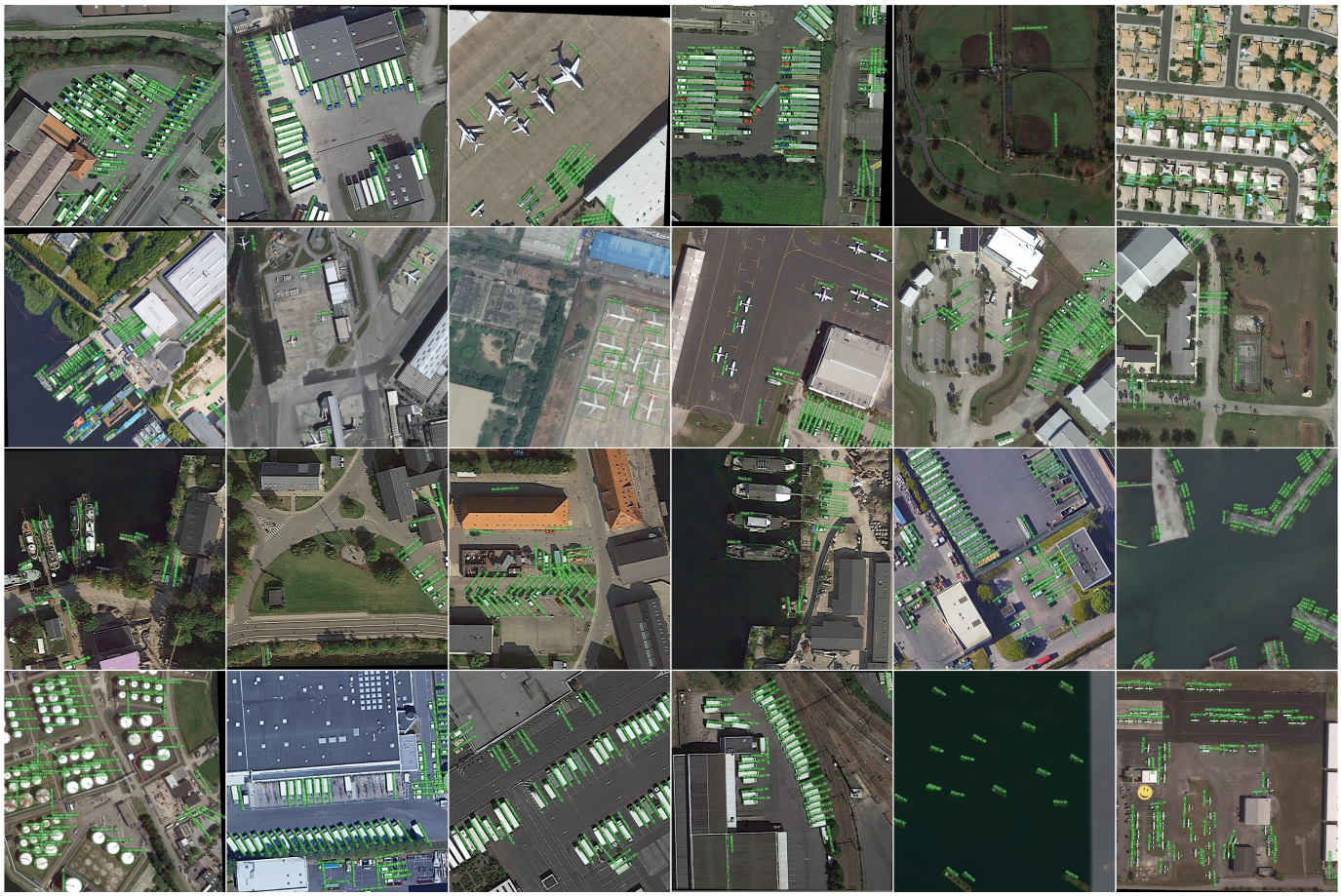


Fig. 7. Some visual detection results. The first two rows are the results of ResNet-50 + UniRS on the MOTA val set, while the remained rows are the results of InternImage-XL + UniRS on the MOTA val set.

TABLE VI
ABLATION STUDIES OF THE NUMBERS AND COMBINATIONS OF TASKS

CLS	RD	SS	***UniRS***	mF_1	mAP	mIoU	Δ_m
✓	-	-	-	80.32	-	-	-
-	✓	-	-	-	73.32	-	-
-	-	✓	-	-	-	63.18	-
✓	✓	-	-	81.02	73.41	-	0.5
✓	✓	-	✓	81.11	73.75	-	0.8
✓	-	✓	-	80.37	-	64.32	0.9
✓	-	✓	✓	81.24	-	64.97	2.0
-	✓	✓	-	-	73.81	64.54	1.4
-	✓	✓	✓	-	74.11	64.91	1.9
✓	✓	✓	-	80.98	73.54	64.45	1.0
✓	✓	✓	✓	82.54	73.91	65.45	2.4

2) Effectiveness of the Proposed Components: We conduct an ablation analysis on the number of activated experts, with the results presented in Fig. 6. We evaluate the performance using the top 4–16 experts to generate task-specific features and plot the accuracy for each task. The results indicate that the proposed model achieves optimal performance when $k = 8$ to $k = 10$. When the value of k is too low, the number of experts involved in feature decoupling is insufficient, failing to effectively expand the feature space. Conversely, when the value of k is too high, involving too many experts in feature assembly leads to a decline in performance.

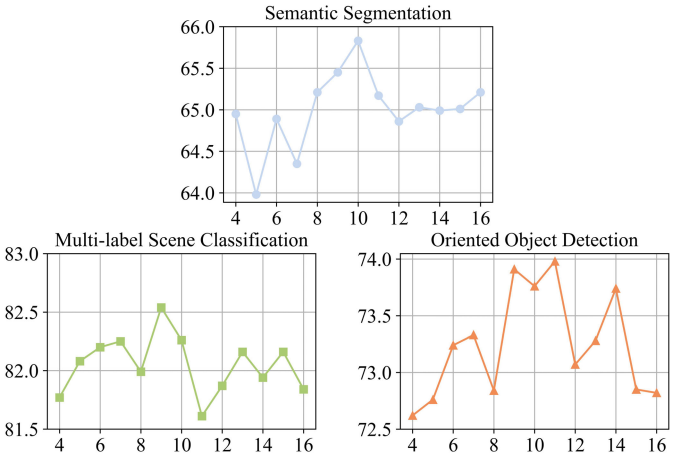


Fig. 8. Ablation study on the number of activated experts k.

3) Task Composition and Cardinality: The interaction between different tasks and the selection of the number of tasks can significantly affect the model's generalization ability and performance. A series of experiments was designed to evaluate the performance of the proposed ***UniRS*** under different task combinations shown in Table VI.

1) *Two-Task Combinations:* The RD + SS pairing achieves the most substantial improvement, with $\Delta_m = 1.4$ without ***UniRS*** and further increasing to $\Delta_m = 1.9$ when

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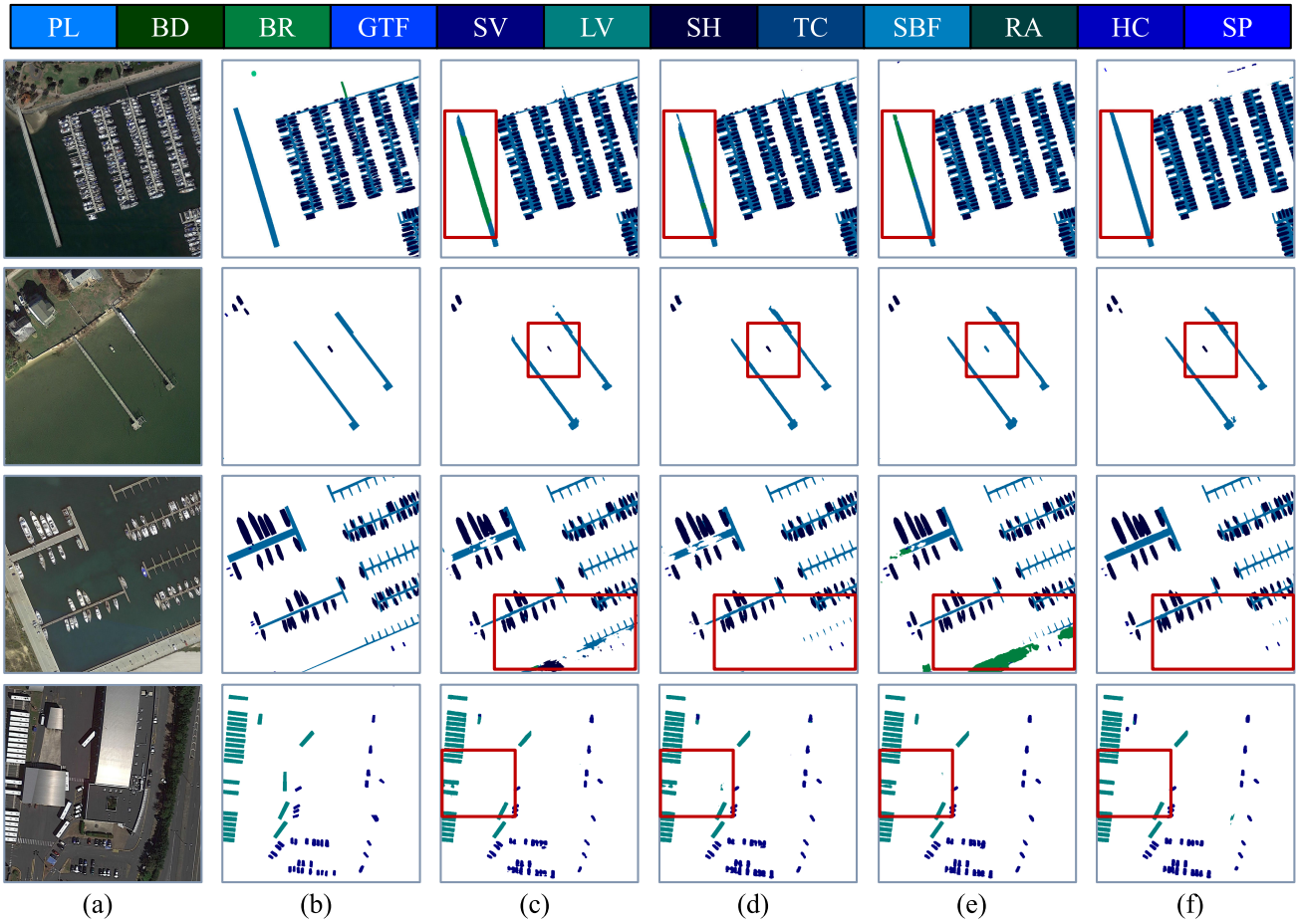


Fig. 9. Visualization of the SS predictions on MOTA validation set. (a) Origin image. (b) Ground Truth. (c) UperNet + ResNet-50. (d) UperNet + ResNet-50 + UniRS. (e) UperNet + ViT-B + RVSA. (f) UperNet + InternImage-XL + UniRS.

applying ***UniRS***. This suggests that detection and segmentation share complementary features, allowing for effective joint learning. The CLS + SS combination results in moderate improvements, particularly with a notable enhancement in segmentation performance when paired with CLS. The Δ_m score rises from 0.9 to 2.0 when using ***UniRS***, demonstrating its ability to enhance synergy between these tasks. CLS + RD provides only marginal performance gains (Δ_m improving from 0.5 to 0.8 with ***UniRS***), likely reflecting the significant difference between CLS and detection tasks, which limits the extent of knowledge transfer.

2) *Three-Task Combination:*| When training all three tasks jointly without ***UniRS***, the improvement remains modest ($\Delta_m = 1.0$), suggesting that simultaneous learning of CLS, detection, and segmentation may introduce interference. However, the application of ***UniRS*** significantly boosts the performance across all tasks, leading to a final $\Delta_m = 2.4$, demonstrating its capability to mitigate negative task interactions and improve overall MTL efficiency.

F. Qualitative Study

1) Visualized Comparison of the Results: We visualize the results of the SS and object detection tasks on the MOTA validation set, as shown in Figs. 7 and 8. It is evident that ResNet-50 trained with ***UniRS*** can more effectively segment targets in SS tasks.

2) Visualization of the Task-Specific Features: We employ Swin-T for multitask fine-tuning on the MOTA dataset and visualize the intermediate feature maps, as shown in Fig. 9. The term "init_feat" denotes the features input into different encoders without using ***UniRS***. In contrast, "cls_feat," "rd_feat," and "ss_feat" represent the features input into the CLS, detection, and segmentation decoders, respectively, after decoupling the generalized features with ***UniRS*** to obtain task-specific features. It is observed that without training using ***UniRS***, the model's focus points are quite disorganized. However, with the application of ***UniRS***, the task-specific features concentrate more on the regions pertinent to solving the specific tasks. For instance, for different fine-grained tasks, ***UniRS*** effectively guides the model to generate more useful features. Features for CLS focus more on broader areas, while segmentation features exhibit better sensitivity to object boundaries. This demonstrates that ***UniRS*** efficiently learns representative generalized features with distinct activation distributions, selectively emphasizing different aspects of the image. Consequently, the gating scores exhibit varying response intensities across different regions, facilitating the selective assembly of experts' features to decode task-specific features.

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V. Conclusion

In this work, we revisit the pretraining and fine-tuning paradigm for remote sensing foundation models, emphasizing a multitask fine-tuning approach. By leveraging the synergies among multiple tasks and generating shared representations, multitask fine-tuning reduces redundant computation, improves generalization, and enhances the overall performance of remote sensing foundation models across various downstream applications. Moreover, we propose ***UniRS***, a multitask fine-tuning framework for remote sensing foundation models that enhances their ability to adapt to various downstream tasks. This framework harnesses the complementary relationships between tasks and facilitates the sharing and interaction of domain knowledge across them. The core components of ***UniRS*** are the CGFI module and the TLMoE decomposition module. CGFI integrates and interacts with multiscale features from the backbone network to produce more generic feature representations. In contrast, the TLMoE module dynamically generates task-specific features by utilizing a mixture of low-rank experts, thereby reducing computational overhead. To evaluate ***UniRS***, we developed the MOTA dataset, which contains over 15 000 remote sensing images with annotations for SS, rotated object detection, and multilabel scene CLS. The experimental results demonstrate that multitask fine-tuning with ***UniRS*** on the MOTA dataset significantly enhances performance across all tasks compared to task-specific fine-tuning.

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